Computer Science Tutorial — ENS Year 1 (EN)

Instructor: BELACEL Madani

Level: ENS — Year 1

Duration: 1h30 (90 min)

Tutorial objective

Practice architecture, OS, algorithms, networks, security, and basic Python.

Exercise 1 — Architecture and encoding *(15 min)*

Task: Explain the role of ALU, control unit, and main memory. Convert `13` to binary.

Detailed answer key:

• ALU: arithmetic and logic operations.

• Control unit: orchestrates fetch-decode-execute.

• Memory: stores instructions and data.

• 13□□ = 1101□ (successive division by 2).

Exercise 2 — Processes and threads *(18 min)*

Task: Distinguish process and thread. Example: web browser.

Detailed answer key:

• Process: own memory space, PID.

• Thread: execution unit sharing process memory.

• Browser: 1 process, UI, rendering, network, extension threads.

Exercise 3 — Maximum algorithm *(18 min)*

Task: Pseudocode for array maximum. Complexity?

Detailed answer key:

max ← T[0]

for i from 1 to n-1 do

if T[i] > max then max ← T[i]

return max

Complexity: O(n) — single pass.

Exercise 4 — Networks *(18 min)*

Task: Differentiate IP, subnet mask, gateway. Role of DNS.

Detailed answer key:

• IP: host identifier.

• Mask: separates network/host parts.

• Gateway: exit to other networks.

• DNS: name → IP address resolution.

Exercise 5 — Python and security *(16 min)*

Task: Write a Python script computing the average of 5 grades. List 3 security best practices.

Detailed answer key:

grades = [12, 14, 11, 15, 13]

print(sum(grades) / len(grades))

Security: MFA, updates, backups, unique passwords.

Session schedule — 90 min

Ex.	Duration	Cumulative
1	15 min	15 min
2	18 min	33 min
3	18 min	51 min
4	18 min	69 min
5	16 min	85 min
Closing	5 min	90 min